Template Document

Academic Support Agent

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Abstract

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1 Insert A head here

This demo file is intended to serve as a “starter file” for ptephy journal papers produced under LaTeX using `ptephy_v1.cls` v0.1

1.1 Insert B head here

Subsection text here.

1.1.1 Insert C head here

Subsubsection text here.

2 Equations

Sample equations.

$$\begin{aligned} \frac{\partial u(t,x)}{\partial t} &= Au(t,x)\left(1-\frac{u(t,x)}{K}\right) - B\frac{u(t-\tau,x)w(t,x)}{1+Eu(t-\tau,x)}, \\ \frac{\partial w(t,x)}{\partial t} &= \delta\frac{\partial^2 w(t,x)}{\partial x^2} - Cw(t,x) + D\frac{u(t-\tau,x)w(t,x)}{1+Eu(t-\tau,x)}, \end{aligned} \tag{1}$$

$$\begin{aligned} \frac{dU}{dt} &= \alpha U(t)(\gamma - U(t)) - \frac{U(t-\tau)W(t)}{1+U(t-\tau)}, \\ \frac{dW}{dt} &= -W(t) + \beta\frac{U(t-\tau)W(t)}{1+U(t-\tau)}. \end{aligned} \tag{2}$$

$$\frac{\partial(F_1,F_2)}{\partial(c,\omega)}_{(c_0,\omega_0)} = \left|\begin{matrix} \frac{\partial F_1}{\partial c} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial c} & \frac{\partial F_2}{\partial \omega} \end{matrix}\right|_{(c_0,\omega_0)} = -4c_0q\omega_0 - 4c_0\omega_0p^2 = -4c_0\omega_0(q+p^2) > 0.$$

3 Enunciations

Assume that $\alpha > 0, \gamma > 1, \beta > \frac{\gamma+1}{\gamma-1}$. Then there exists a small $\tau_1 > 0$, such that for $\tau \in [0, \tau_1)$, if c crosses $c(\tau)$ from the direction of to a small amplitude periodic traveling wave solution of (2.1), and the period of $(\breve{u}^p(s), \breve{w}^p(s))$ is

$$\check{T}(c) = c \cdot \left[\frac{2\pi}{\omega(\tau)} + O(c - c(\tau)) \right].$$

From (0.8) and (2.10), it holds $\frac{d\omega}{d\tau} < 0$, $\frac{dc}{d\tau} < 0$ for $\tau \in [0, \tau_1)$. This fact yields that the system (2.1) with delay $\tau > 0$ has the periodic traveling waves for smaller wave speed c than that the system (2.1) with $\tau = 0$ does. That is, the delay perturbation stimulates an early occurrence of the traveling waves.

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4 Figures & Tables

The output for figure is:

Figure 1: Insert figure caption here

An example of a double column floating figure using two subfigures. (The subfig.sty package must be loaded for this to work.) The subfigure `\label` commands are set within each subfloat command, the `\label` for the overall figure must come after `\caption`. `\hfil` must be used as a separator to get equal spacing. The subfigure.sty package works much the same way, except `\subfigure` is used instead of `\subfloat`.

The output for table is:

Table 1: An Example of a Table.

One	Two
Three	Four

5 Conclusion

The conclusion text goes here.

Acknowledgment

Insert the Acknowledgment text here.

For references, note how to include DOI information from examples below.

References

[1] J. P. Blaizot, and E. Iancu, Phys. Rep. **359**, 355 (2002). https://doi.org/10.1016/S0370-1573(01)00061-8

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[4] S. Alekhin, A. Djouadi, and S. Moch, Phys. Lett. B 716, 214 (2012) [arXiv:1207.0980 [hep-ph]]. https://doi.org/10.1016/j.physletb.2012.08.024

6 Appendix head

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$$a + b = c \tag{3}$$

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Table 2: An Example of a Table.

One	Two
Three	Four

6.1 Appendix subhead

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$$a + b = c \tag{4}$$

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Figure 2: Figure caption

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$$a + b = c \tag{5}$$

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7 Appendix heading

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$$a + b = c \tag{6}$$

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$$a + b = c \tag{7}$$

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Figure 3: Figure caption

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Figure 4: Figure caption

Table 3: An Example of a Table.

One	Two
Three	Four

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